SafeHer

Women Safety Analytics Dashboard

Empowering women's safety through smart insights and informed choices.

Project Report

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AI -ML VITYARTHI PROJECT

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1. What is SafeHer?

So, SafeHer is basically a dashboard I built to help look at crime data related to women in India and also predict how safe a person might be in a given situation. I wanted to make something that actually feels useful, not just a project that shows some graphs and calls it a day.

There are two main parts to it. The first is a Crime Analysis Dashboard that uses real data from the NCRB that's the National Crime Records Bureau going all the way from 2001 to 2021. It covers 7 types of crimes across 20+ Indian states. The second part is an AI-powered Risk Predictor where you can describe your current situation (like what time it is, where you are, whether you're alone), and it tells you if you're in a Low, Medium, or High risk situation. There's also a third tab that shows how well the model is actually performing, which I thought was important to include so users know what they're trusting.

2. The Problem I Wanted to Solve

2.1 Why this matters

Honestly, this topic feels important on a very personal level. It's not just "data" or "statistics" .It's something that affects real people around us every day. We all hear about cases in the news or from people we know, but most of the time it just stays as information and doesn't translate into something actionable.

What really stood out to me was that even though so much data exists (especially from NCRB), it's not something an average person can easily understand or use. Like, no one is going to open a 50-page PDF or an Excel sheet before deciding whether a place is safe or not. So even though the data is technically "available," it's not actually accessible.

Also, safety decisions are usually made in the moment , like whether to take a certain route, step out at night, or travel alone. In those situations, people rely more on instinct than actual data, because there isn't any simple way to get a quick insight.

That's where I felt this project could actually matter ,by turning raw data into something understandable and combining it with a predictive model, it can help people make slightly more informed decisions instead of just guessing.

I thought, what if we could take this data and present it in a way that's actually easy to understand? And on top of that, what if we could use machine learning to give someone a quick heads-up about how safe their current situation is?

2.2 The gap I noticed

While working on this, I realized that the problem isn't just the lack of data it's more about how that data is (not) being used.

First, there's a clear gap in accessible safety information. Even though crime data exists, it's not presented in a way that helps someone decide something simple like "Is this area generally safe?" or "Has this type of crime increased recently?" There's a disconnect between data availability and usability.

Second, most existing safety apps are reactive. They focus on emergency situations like calling for help after something has already gone wrong. While that's obviously important, there's very little focus on prevention. There aren't many tools that try to answer: "Am I entering a risky situation right now?"

Another gap I noticed is that situational factors are ignored. Safety isn't just about location it depends on multiple things like time, lighting, whether someone is alone, etc. But most platforms don't combine these factors to give a more realistic picture of risk.

Finally, from a technical point of view, I felt like machine learning is underused in this space. We use ML for recommendations, ads, and predictions everywhere so why not use it to model something like safety risk? Even a basic model can capture patterns between different factors and give a useful output.

2.3 Scope

On one side, the crime analysis dashboard focuses on real NCRB data across 20+ states and multiple crime categories over 21 years. The idea here is not just to show charts, but to actually highlight patterns like which states consistently report higher cases, how trends have changed over time, and which types of crimes are more common.

On the other side, the risk predictor is more personal and situation-based. Instead of relying on large-scale historical data alone, it focuses on factors that a person can observe in real time ,like whether it's night, if the area is isolated, lighting conditions, etc. Based on this, it gives a simple output (Low, Medium, High risk), which is easy to understand.

Of course, there are limitations. The predictor uses synthetic data because real-world labeled data for personal safety situations isn't publicly available. So the model isn't perfect, but it still works as a proof of concept for how this idea could be developed further.

Overall, the scope is not to build a fully accurate safety system, but to create something that:

- Makes crime data easier to understand
- Shows how ML can be applied to real-life problems
- Gives users at least some level of awareness about their situation

3. The Dataset

3.1 The real NCRB crime data

Property	Detail
Dataset name	Crimes Against Women in India
Source	National Crime Records Bureau (NCRB) via Kaggle
Coverage	2001 to 2021 (21 years)
States covered	20+ major Indian states and Union Territories
Crime categories	Rape, Kidnapping & Abduction, Dowry Deaths, Assault on Women, Assault on Modesty, Domestic Violence, Women Trafficking
File	CrimesOnWomenData.csv (736 rows x 10 columns)

3.2 Cleaning the data

Honestly, cleaning this dataset took longer than I expected. A few things I had to fix:

- The state names were in UPPERCASE for years 2001–2010 and in Title Case for 2011–2021. So Uttar Pradesh and UTTAR PRADESH were showing up as two different states. Fixed this with str.title().
- Some state names had weird variations like 'Delhi UT' or 'D&N Haveli' that I had to standardize manually.
- Removed some really small Union Territories since their numbers were so low they were just cluttering the charts.
- Converted the data from wide format (one column per crime) to long format (one row per state-year-crime combo) using pd.melt() so it was easier to work with.

3.3 The synthetic risk data

There's no public dataset that tracks personal situational safety, so I had to create one. I generated 3,000 synthetic records using a weighted scoring formula based on known risk factors. For example, being out at night adds more to the risk score than being in a well-lit area reduces it. I also added some Gaussian noise to make it feel more realistic. I know synthetic data has limitations, but there really wasn't another option here.

4. How the App is Structured

The whole app runs from just four files:

File	What it does
app.py	Everything — the data loading, the ML pipeline, and all the UI in one file
CrimesOnWomenData.csv	The real NCRB crime data used for the dashboard
requirements.txt	Lists the 6 Python libraries needed to run the app
README.md	Setup guide for anyone who wants to run it locally

The app has three tabs:

Tab	Name	What data it uses
Tab 1	Crime Analysis Dashboard	Real NCRB data from CrimesOnWomenData.csv
Tab 2	AI Risk Predictor	Synthetic domain-informed data (no public dataset exists for this)
Tab 3	Model Performance	Evaluation metrics from the trained Random Forest model

5. How I Built It

5.1 Data processing

- Load the CSV with pandas
- Normalize state names to fix the casing issue
- Melt the wide-format data into long format so groupby operations work cleanly
- Filter to major states so the visualizations are actually readable

5.2 The ML pipeline

Step	What I did
Feature engineering	Encoded all categorical columns using sklearn LabelEncoder; saved the encoder objects for inference
Train/test split	80/20 stratified split — 2,400 training samples, 600 test samples
Model	RandomForestClassifier with 100 trees, class_weight='balanced', random_state=42
Caching	@st.cache_resource so the model only trains once per session, not on every click
Evaluation	Accuracy, precision/recall/F1 per class, confusion matrix, and feature importance

5.3 Visualization

All the charts use Matplotlib and Seaborn with a consistent pink-and-maroon color palette .I used horizontal bar charts for state comparisons and feature importance, a pie chart for crime type breakdown, a line chart with fill for the 21-year trend, seaborn heatmaps for the confusion matrix and state×crime breakdown, and confidence bars for the risk predictor output.

6. Design Decisions I Made

Decision	Why I made it
Real NCRB data for crime analysis	Using official government data makes the dashboard actually credible. Anyone can verify the numbers.
Synthetic data for the risk predictor	No labelled dataset exists for personal situational safety. Domain-informed synthetic data is the standard workaround.
Single-file app (app.py)	Makes it way easier to share and run. No imports between modules, no confusion about file structure.

Random Forest classifier	Handles mixed feature types naturally, gives feature importance scores, and worked better than simpler models on this data.
class_weight='balanced'	Without this, the model just predicts High all the time because it's the majority class. This forces it to learn all three classes properly.
Wide to long data format	Makes all six chart types work with simple groupby calls on one consistent dataframe.
Emergency helplines in the UI	Felt important to close the loop from 'here's your risk level' to 'here's what you can do about it'.

7. Challenges I Ran Into

7.1 The state name casing issue

This was the most annoying bug to track down. The NCRB dataset uses UPPERCASE state names for 2001–2010 and Title Case for 2011–2021. So when I grouped by state, Uttar Pradesh and UTTAR PRADESH were showing up as separate entries and the total counts were being split. Fixing it with str.title().str.strip() and a replacement dictionary for edge cases like D&N Haveli sorted it out.

7.2 Wide vs long format

The original CSV has one column per crime type, which sounds fine until you try to make charts or do groupby operations. I needed to melt it into long format , one row per state-year-crime combination , before any of the visualizations would work cleanly.

7.3 Class imbalance

The synthetic dataset generates more High-risk records by design (which makes sense for a safety tool), but it meant the model was just predicting High for everything. class_weight='balanced' and stratify=y in train_test_split fixed this. Low-risk recall ended up at 0.65 which is a bit lower, but for a safety tool it's better to over-predict risk than under-predict it.

7.4 Streamlit performance

Without caching, the model was retraining and the CSV was reloading every time I touched a widget. That made the app basically unusable. Wrapping load_crime_data() in @st.cache_data and train_model() in @st.cache_resource meant both run exactly once per session.

7.5 Label encoding at prediction time

The LabelEncoder mappings from training have to be reused at inference .You can't just create a new encoder and hope the mappings match. I solved this by returning the le_dict from the cached train_model() function and passing it into the prediction step.

8. Results

8.1 What the crime data showed

- Uttar Pradesh has the highest cumulative case count across all 21 years, which lines up with what NCRB reports say.
- Domestic Violence and Assault on Women are the most common crime categories.
- All categories show an upward trend from 2001 to 2021 ,this reflects both more incidents and better reporting.
- Madhya Pradesh, West Bengal, and Rajasthan are consistently in the top 4.
- There's a visible dip around 2020 on the trend line, almost certainly because of COVID-19 lockdowns reducing reported cases.

8.2 Model performance

Metric	Low	Medium	High
Precision	0.72	0.66	0.92
Recall	0.65	0.76	0.86
F1-Score	0.68	0.71	0.89
Test samples	74	189	337

Overall accuracy: 80.33%. The model does best on the High-risk class, which is what you want for a safety tool — it's correct 92% of the time when it predicts High, and catches 86% of actual High-risk cases.

8.3 What features matter most

- Location Type - most important overall; being on an isolated road is the single biggest risk factor
- Time of Day -night substantially increases predicted risk
- Travelling Alone - strong risk amplifier
- Lighting Condition -dark environments significantly increase risk
- Past Incidents Nearby, Familiar Area, CCTV Present, Phone Charged — moderate contributors

9. What I Learned

9.1 Technical things learnt

- How to properly clean and reshape real government CSV data, the wide-to-long transformation and casing normalization were both things I hadn't done at scale before.
- How to structure a multi-tab Streamlit app with proper caching for both data and model objects.
- Practical approaches to handling class imbalance in supervised classification.
- How to surface ML interpretability (feature importance, confusion matrix) in a way that makes sense to non-technical users.
- How to write a requirements.txt with minimum version pins for reproducibility.

9.2 The Bigger picture:

- Real data is genuinely more powerful than synthetic data the COVID-19 dip in 2020 on the trend line is something a synthetic dataset could never have captured.
- Even official government data needs significant cleaning. The casing inconsistency in the NCRB dataset wasn't obvious until I looked at the grouped output.
- ML can actually serve a social purpose if the problem is framed right converting raw statistics into a personal, real-time safety signal felt genuinely useful to build.
- Being upfront about data limitations (especially using synthetic data for the risk predictor) builds more trust than hiding it.

10. Self-Evaluation

What I'm judging	My honest take

Problem relevance	Women's safety in India is a real, documented problem and this project applies data analysis, supervised ML, and visualization — all core course topics.
Solution quality	The crime dashboard uses real NCRB data which is a meaningful upgrade from a purely synthetic approach. The AI predictor hits 80.33% accuracy and 0.89 F1 on the High-risk class.
Code quality	Single well-commented file with clearly named functions. Data loading, training, and UI are properly separated. All 6 dependencies are pinned.
Depth of reflection	This report covers the real data cleaning issues, the ML challenges (class imbalance, caching, label encoding), and actual findings from the data including the COVID dip in 2020.
README usability	Four files, three commands. The README covers the data source, all three tabs, model details, and emergency helplines.
What I'd do better	The risk predictor still uses synthetic data — I'd try to find or collect real incident data for a future version. I'd also add a choropleth map and deploy it on Streamlit Cloud.

11. What I'd Add Next

- A choropleth map (using Plotly or Folium) for geospatial crime visualization by state — it'd be a lot more intuitive than bar charts for this.
- More recent NCRB data as it gets released (2022 onwards).
- A year-range slider so users can explore specific time periods.
- Deploy on Streamlit Cloud so anyone can use it without installing anything locally.
- Translate the interface into regional Indian languages for wider reach.
- Try to find or collect real-world safety incident data to replace the synthetic risk training set.

12. Run the program

You need four files — all of them are in the GitHub repository:

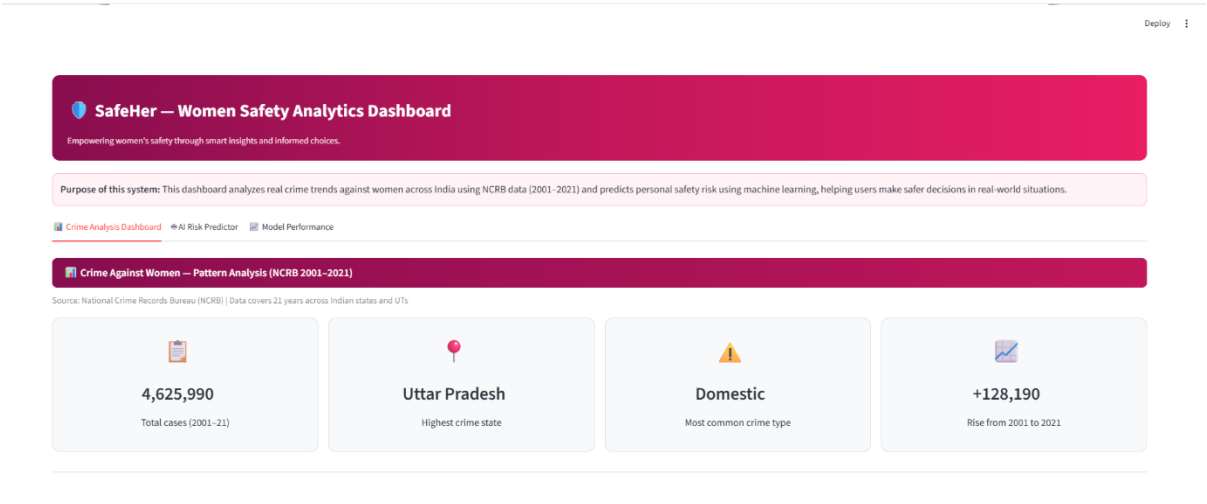
File	Where to get it
app.py	GitHub repository
CrimesOnWomenData.csv	GitHub repository (originally from Kaggle NCRB dataset)
requirements.txt	GitHub repository
README.md	GitHub repository

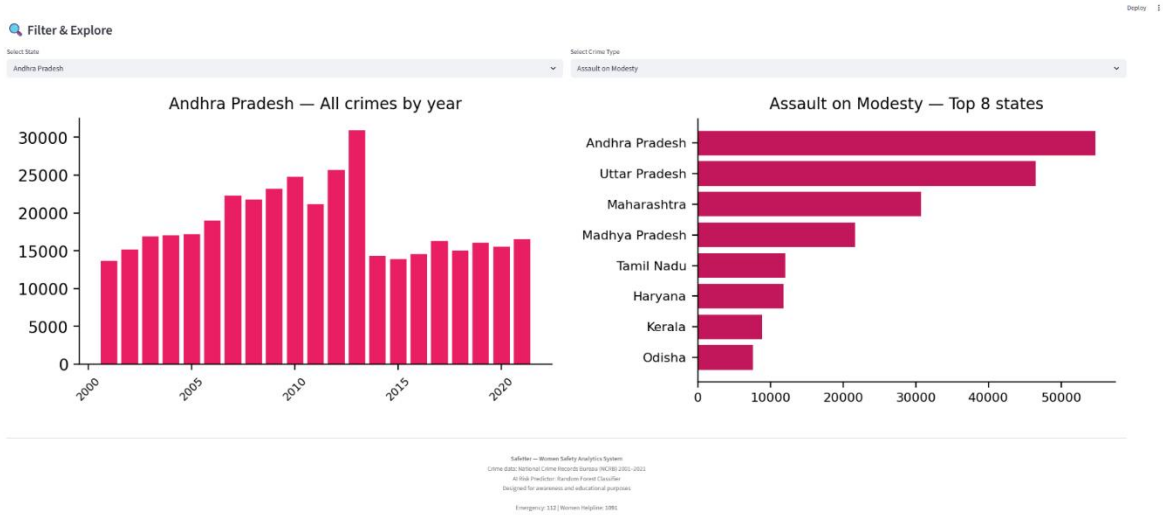
Three commands to get it running:

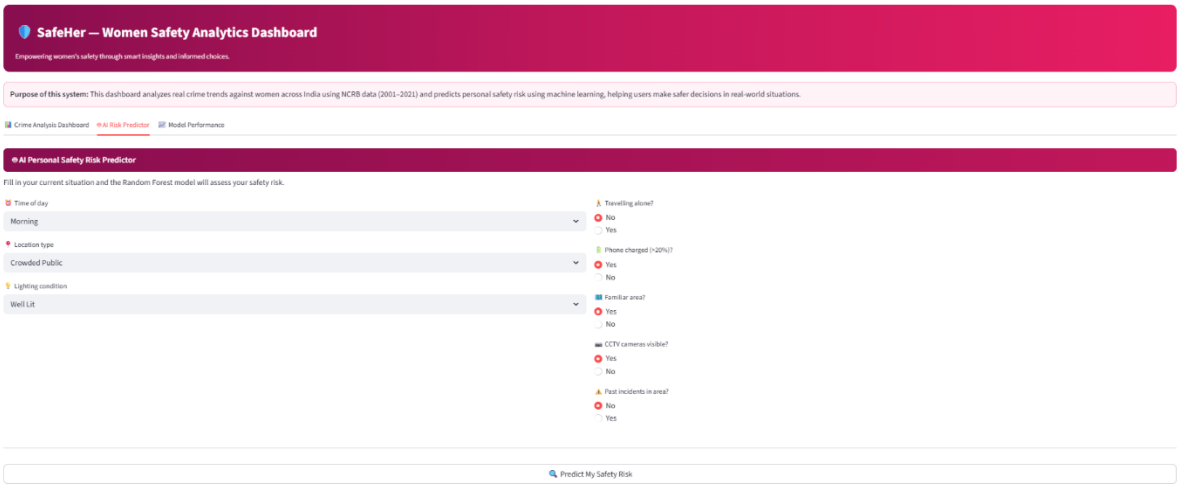
```
git clone https://github.com/suhaniboxi-creator/AI-ML-PROJECT-.git
cd AI-ML-PROJECT-
pip install -r requirements.txt
streamlit run app.py
```

The app opens at http://localhost:8501. Just make sure CrimesOnWomenData.csv is in the same folder as app.py or it won't load.

13. Website Screenshots





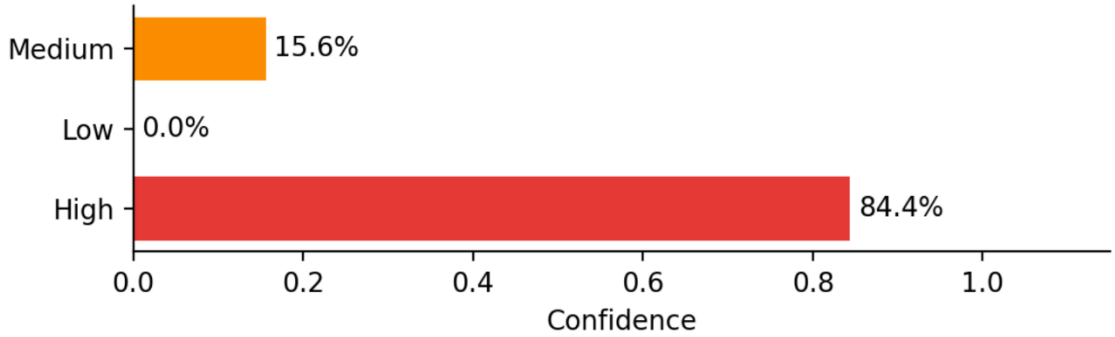


HIGH RISK

Your situation shows strong danger indicators.

- Move to a crowded and well-lit place immediately
- Call a trusted contact and stay connected
- Be ready to dial emergency number 112

Confidence Breakdown



Deploy

SafeHer — Women Safety Analytics Dashboard

Empowering women's safety through smart insights and informed choices.

Purpose of this system: This dashboard analyzes real crime trends against women across India using NCRB data (2001–2021) and predicts personal safety risk using machine learning, helping users make safer decisions in real-world situations.

Crime Analysis Dashboard | AI Risk Predictor | Model Performance

This section evaluates how accurately the machine learning model predicts safety risk.

ML Model Performance & Evaluation

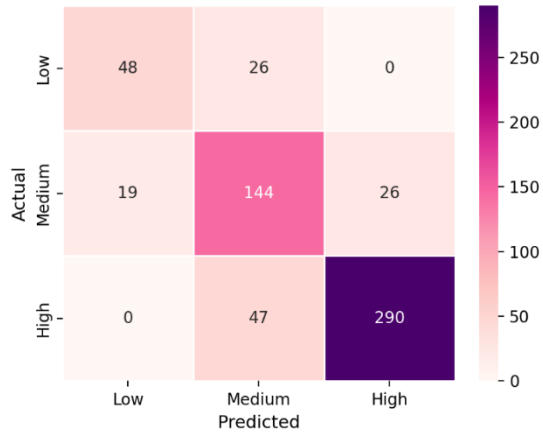
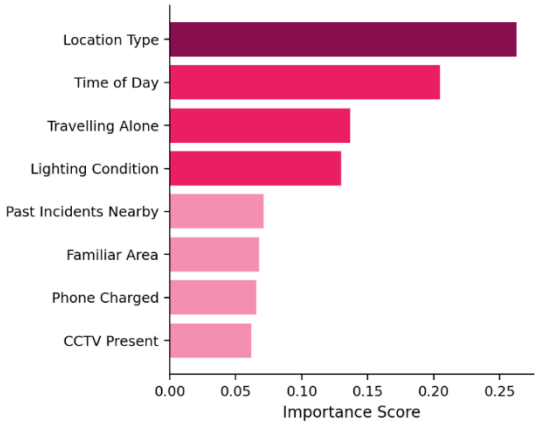
Model: Random Forest Classifier | Training: 2400 | Test: 600 | Accuracy: 80.33%

How the model works:

- Uses Random Forest (ensemble machine learning technique)
- Considers 8 personal safety factors as input features
- Predicts risk as Low, Medium, or High
- Trained on domain-informed synthetic data (no public dataset exists for personal safety situations)

Deploy

Confusion Matrix



Feature Importance